



Energy Management System

PRESENTATION

02 May, 2024



EMS FEATURES AND BENEFITS

User-Friendly Interface:

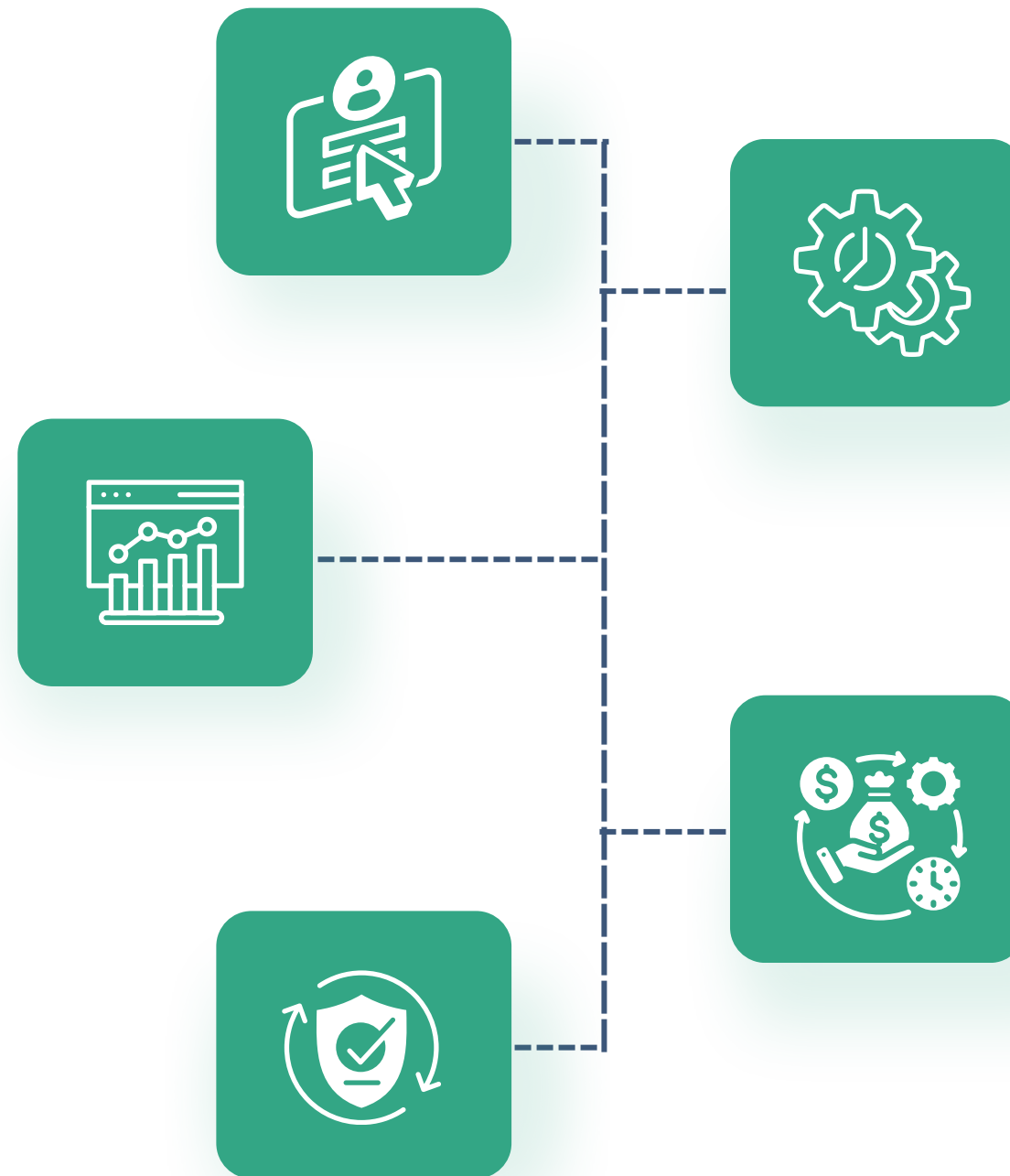
- Interactive dashboards to visualize energy consumption patterns.
- Graphs, charts, and heatmaps to illustrate trends and anomalies.

Data Analytics and Reporting:

- Advanced data analytics for identifying energy-saving opportunities.
- Customizable reporting for detailed energy usage insights.

Security and Compliance:

- Ensures data security and privacy.
- Complies with industry standards and regulations.



Alert and Notification System:

- Real-time alerts for abnormal energy usage, equipment faults, or threshold breaches.
- Notifications alerts to relevant personnel.
- Configurable alert thresholds based on predefined rules.

Cost Management:

- Tracks and analyzes energy costs.
- Identifies cost-saving opportunities and ROI of energy projects.

EMS FEATURES AND BENEFITS

Real-Time Monitoring and Data Acquisition:

- Continuous monitoring of energy usage across equipment and facilities.
- Data collection from IoT sensors, smart meters, and other monitoring devices.

Energy Analytics and Insights

- Advanced analytics to identify inefficiencies and potential energy savings.
- Predictive analysis to forecast energy demand and consumption patterns

User and Role Management

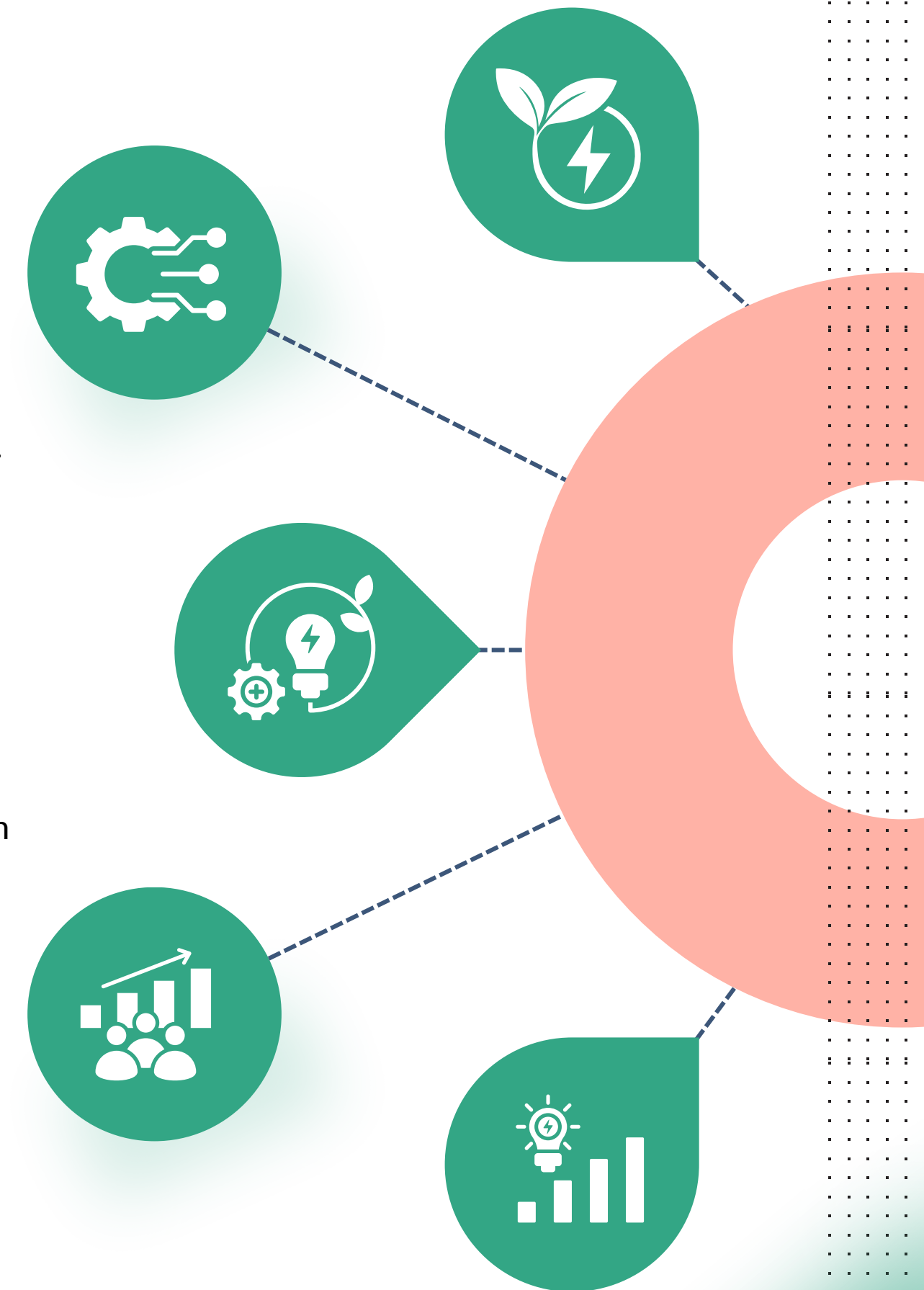
- User authentication and authorization with role-based access control.
- Ability to define different user roles such as admin, operator, and auditor

Predictive Maintenance:

- Predictive analytics to foresee equipment failures.
- Reduces downtime and maintenance costs.

Scalability:

- Scalability to accommodate future expansion or additional facilities.
- Integration with multiple data sources such as gas, electricity, water, and renewables.



ENERGY MANAGEMENT SYSTEM INSIGHTS

Energy



Monitor real-time energy consumption, trends, and detailed reports for effective energy management.

Genset



Tracks generator performance, fuel consumption, and machine analytics to ensure optimal operation.

Water



Provides insights into water usage, source breakdown, and section-wise analysis for efficient water management.

HVAC



Manages heating, ventilation, and air conditioning systems, ensuring optimal environmental conditions and energy efficiency.

Alerts

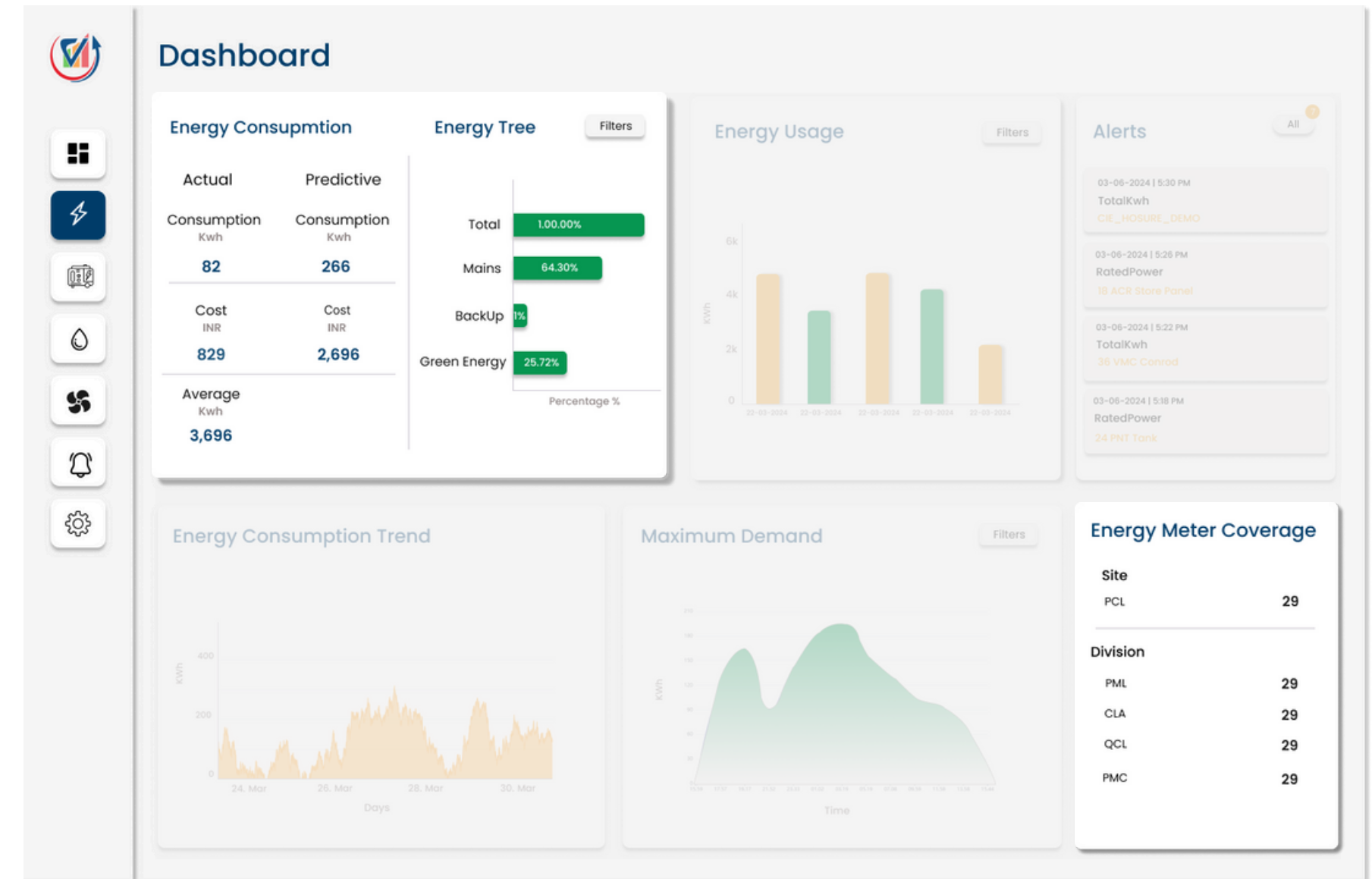


Displays critical alerts for immediate attention to any issues or anomalies in the system.

Energy Dashboard

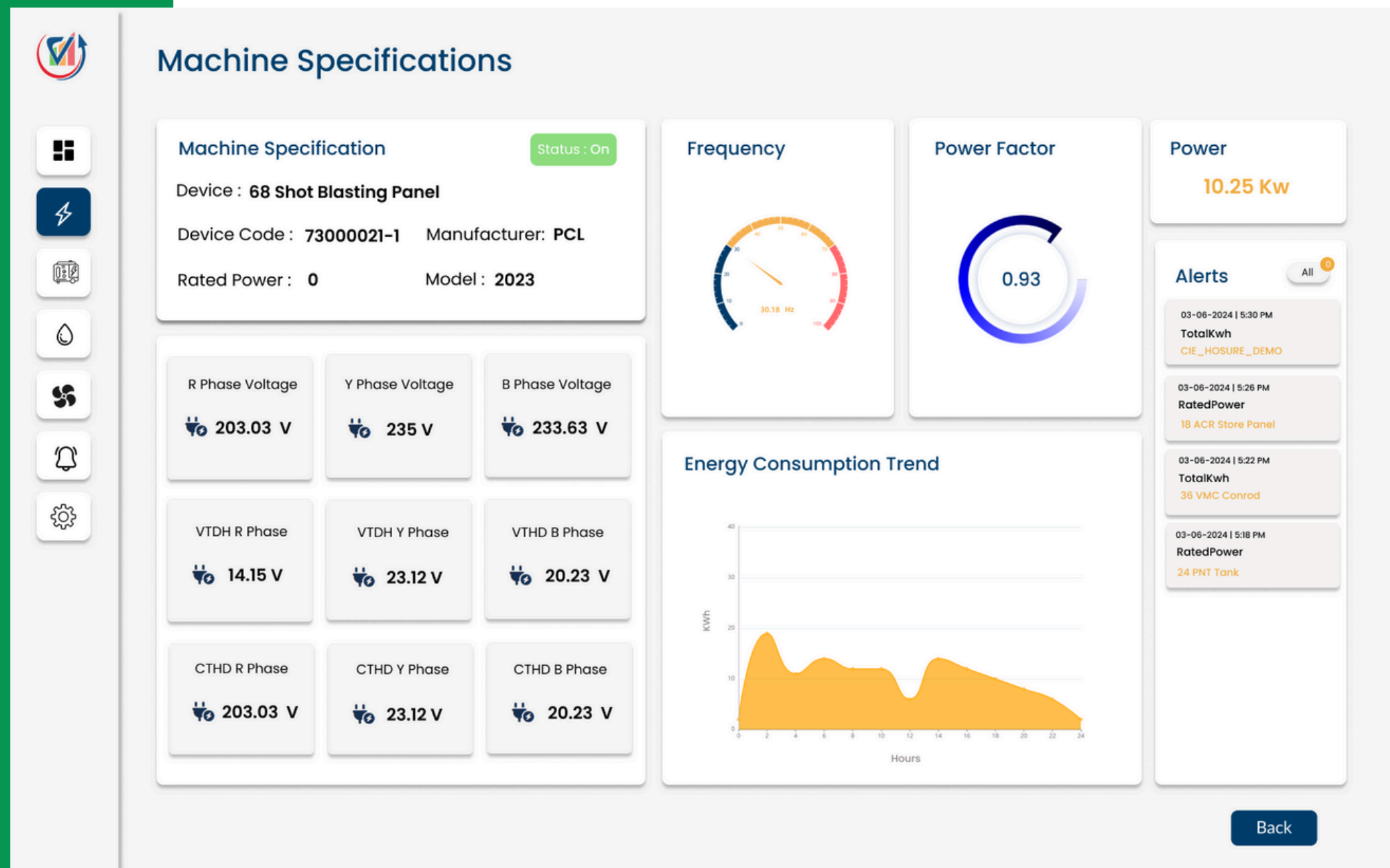
Energy Consumption

- Real-time monitoring of energy consumption
- Filters: daily, yesterday, monthly views
- Comparison of actual vs. predictive energy usage
- Visual representation of contributions from various energy sources
- Track energy supply from each source
- Compare different energy flow scenarios
- Analyze average daily and monthly kWh consumption
- Insights for optimizing energy efficiency



Energy Meter Coverage

- Displays the total number of energy meters connected to the platform
- Provides an overview of the scope and scale of monitored energy assets



Machine Specifications

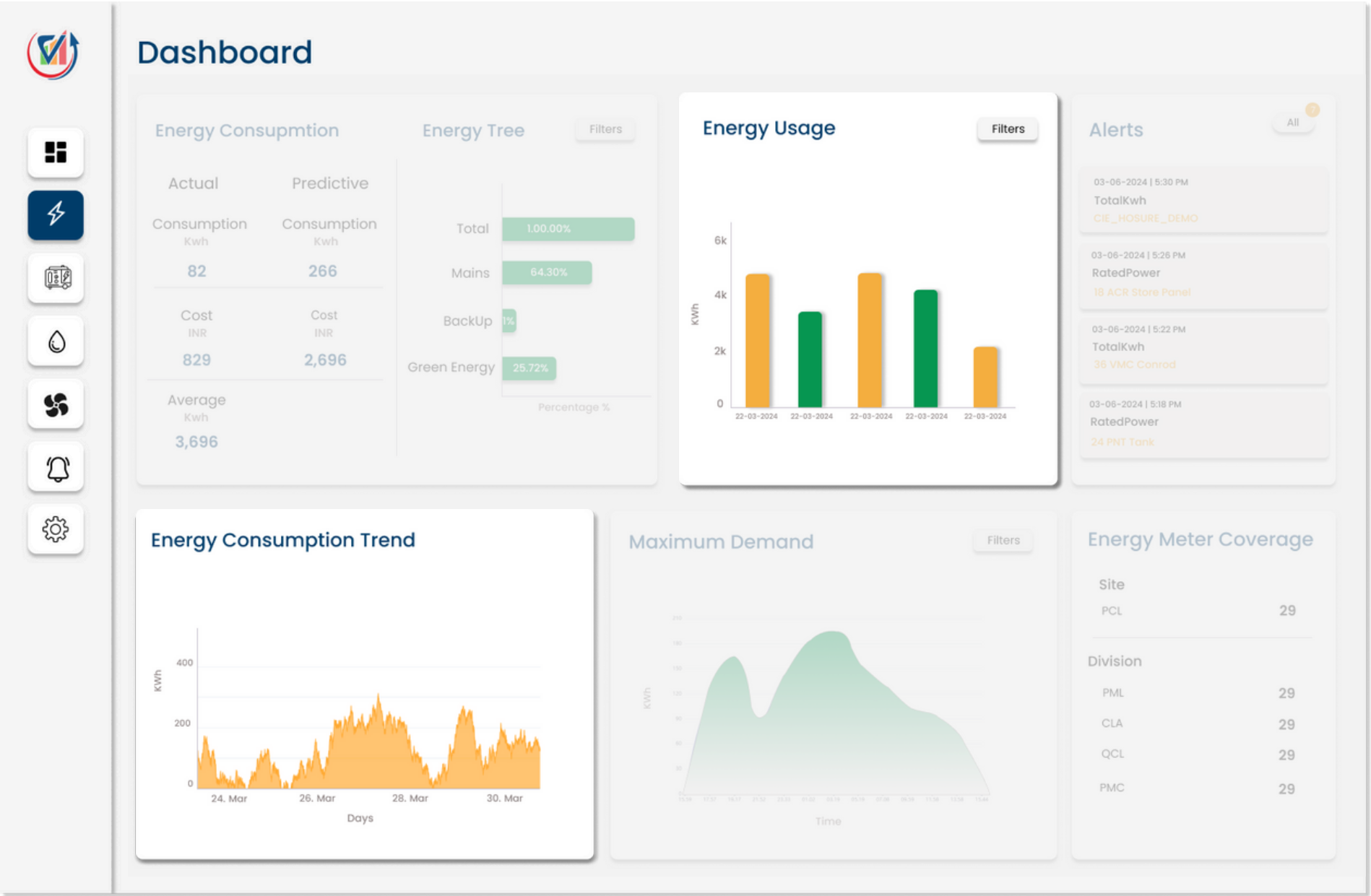
- Individual equipment monitoring
- Click to view essential parameters: frequency, power factor, voltage, current
- Displays energy trends and alerts
- Enables detailed monitoring
- Facilitates proactive management of device performance and efficiency

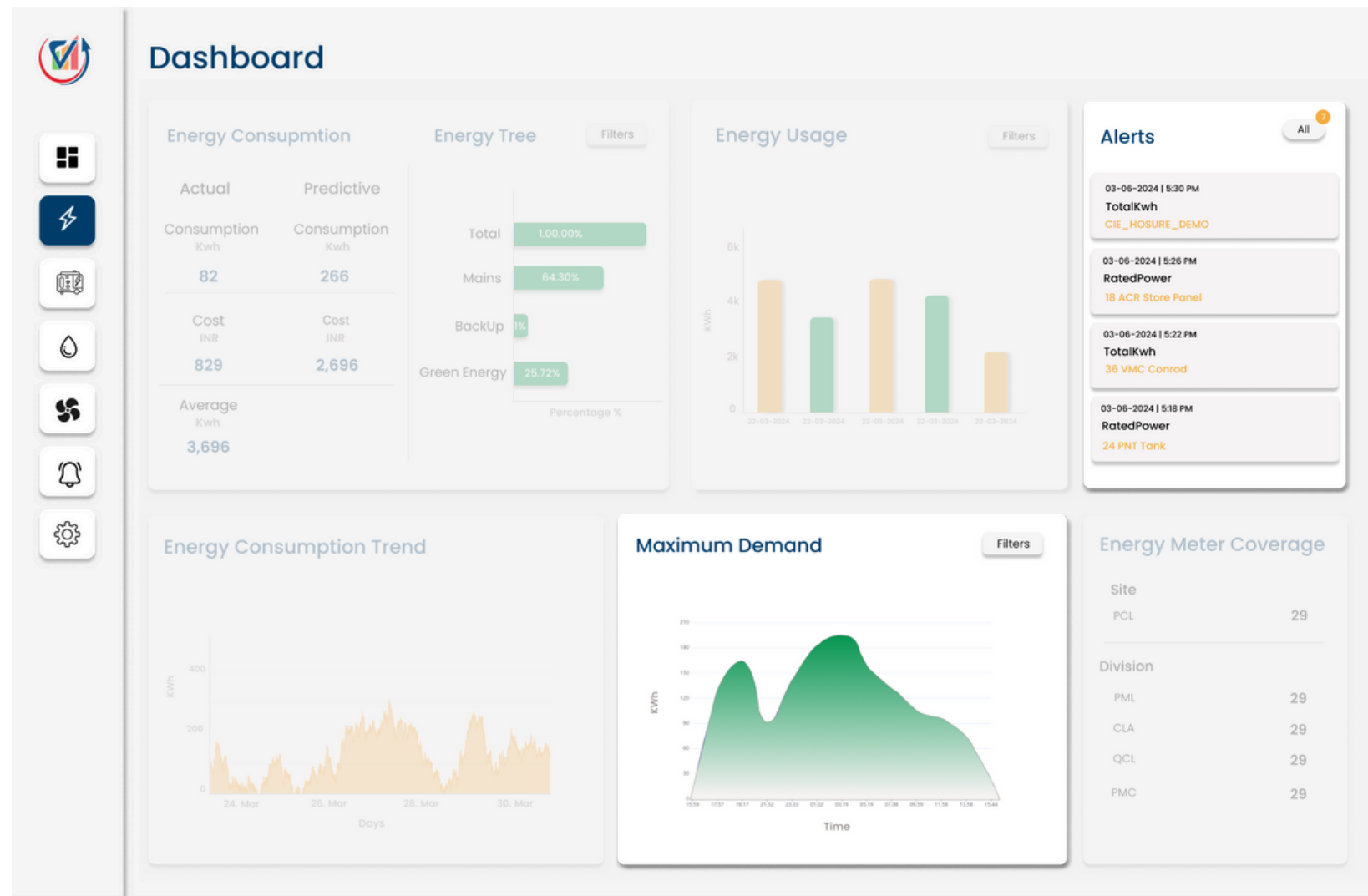
Energy Usage

- Displays energy consumption by month, day, divisions, and sitewise
- Drill-down for detailed monthly and daily breakdowns
- Analyze associated costs
- Provides comprehensive insights

Energy Consumption Trend

- Tracks real-time energy usage
- Provides continuous updates
- Offers insights into current energy consumption patterns





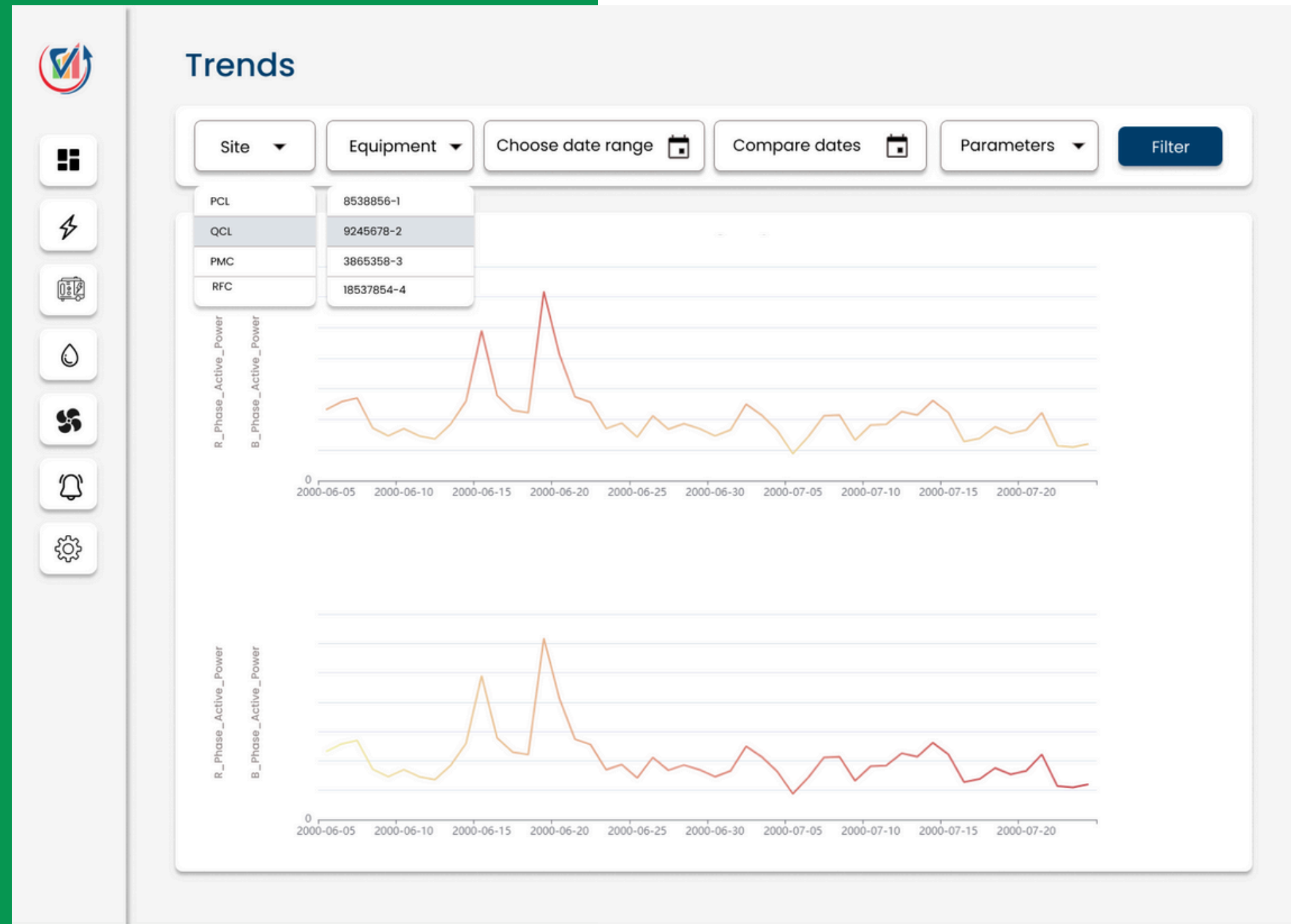
Maximum Demand

- Displays real-time peak power values in kilowatts (kW)
- Visible from mains level and drillable down to individual equipment
- Provides detailed insights into peak power demands across different levels of the energy infrastructure

Alerts

- Notify users of critical events or deviations from normal operating conditions
- Triggered based on predefined thresholds or conditions
- Proactively warn users about:
 - Equipment malfunctions
 - Energy inefficiencies
 - Safety concerns
- Provide timely notifications for effective management and maintenance

Trends



Trends

- Visualizes time-series movements of key parameters in graphical form.
- Allows switching between different parameters.
- Helps spot abnormalities and understand patterns affecting performance.
- Aids in proactive monitoring and optimization of energy system performance.

Reports

Reports provides detailed insights into overall energy consumption & usage, offering comprehensive data and analysis to support informed energy management decisions.

Equipment-wise Daily Consumption

- Displays total units consumed monthly
- Analyzes daily consumption trends for each equipment.
- Provides a detailed view of energy usage patterns over time.

Equipment-wise Daily Cumulative Energy

- Provides meter readings for individual equipment over a period of up to 30 days
- Offers a detailed account of daily energy consumption in table form.

Equipment-wise Monthly Consumption

- This report shows the energy consumption of individual equipment for a selected year.
- It enables analysis and comparison of consumption across different months.
- Provides insights into usage trends and variations over time.



Shield History

- Tracks total energy usage at equipment and machine levels.
- Provides detailed insights using customizable filters and parameters.
- Facilitates comprehensive analysis and historical tracking of energy consumption.

Division-wise Daily Consumption

- Provides detailed breakdown of energy usage across organizational divisions daily.
- Allows analysis and comparison of energy consumption patterns among different segments over time.

Division-wise Monthly Consumption

- Allows analysis of energy usage across plant divisions monthly.
- Provides comprehensive insights into consumption patterns.
- Facilitates informed decision-making for energy management.



Reports

Clear table X

Report TypeShield history

SiteKTL

Equipment73000023-1

Choose date range09/07/2024 - 10/07/2024

Export

Shield History

Date	Equipment	Ideal Kwh	OnLoadKwh	TotalKwh	MinKwh	MaxKwh	Cost
2024-07-09	73000023-01	0	43.12	43.12	0	0	452

- ### Equipment-wise Monthly Consumption
- This report shows the energy consumption of individual equipment for a selected year.
 - It enables analysis and comparison of consumption across different months.
 - Provides insights into usage trends and variations over time.

- ### Shield History
- Tracks total energy usage at equipment and machine levels
 - Provides detailed insights using customizable filters and parameters & Enables data download in Excel format.
 - Facilitates comprehensive analysis and historical tracking of energy consumption.



Division-wise Daily Consumption

- This report provides a detailed breakdown of energy usage across different divisions within an organization on a daily basis.
- It allows analysis and comparison of energy consumption patterns among various segments over time.

Division-wise Monthly Consumption

- These reports allow analysis of energy usage across different divisions within the plant on a monthly basis.
- Provides comprehensive insights into consumption patterns.
- Facilitates informed decision-making for energy management.

Water Dashboard

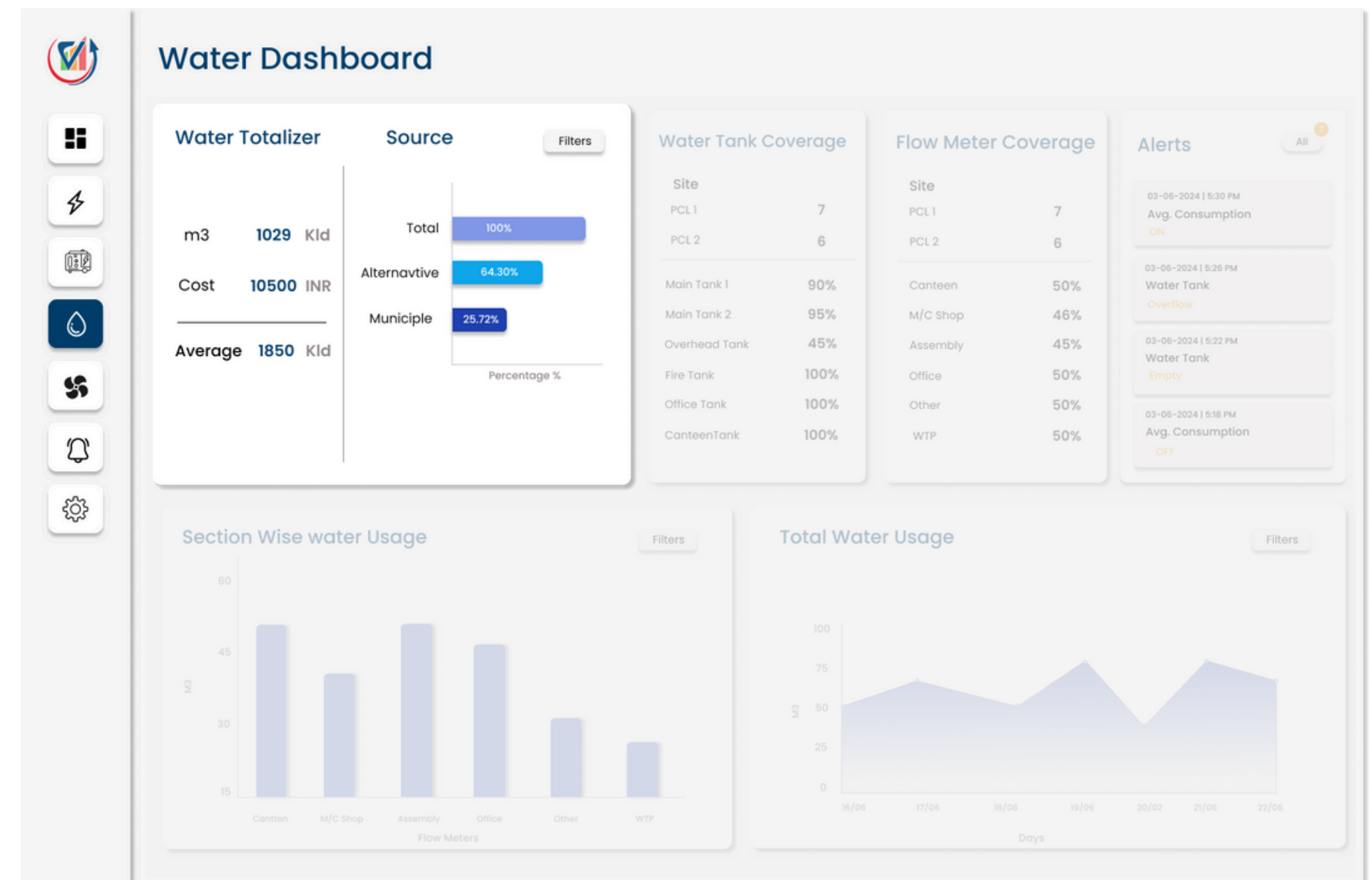
The water dashboard provides a comprehensive overview of water usage, enabling users to monitor consumption, track costs, and optimize resources efficiently for effective management and conservation.

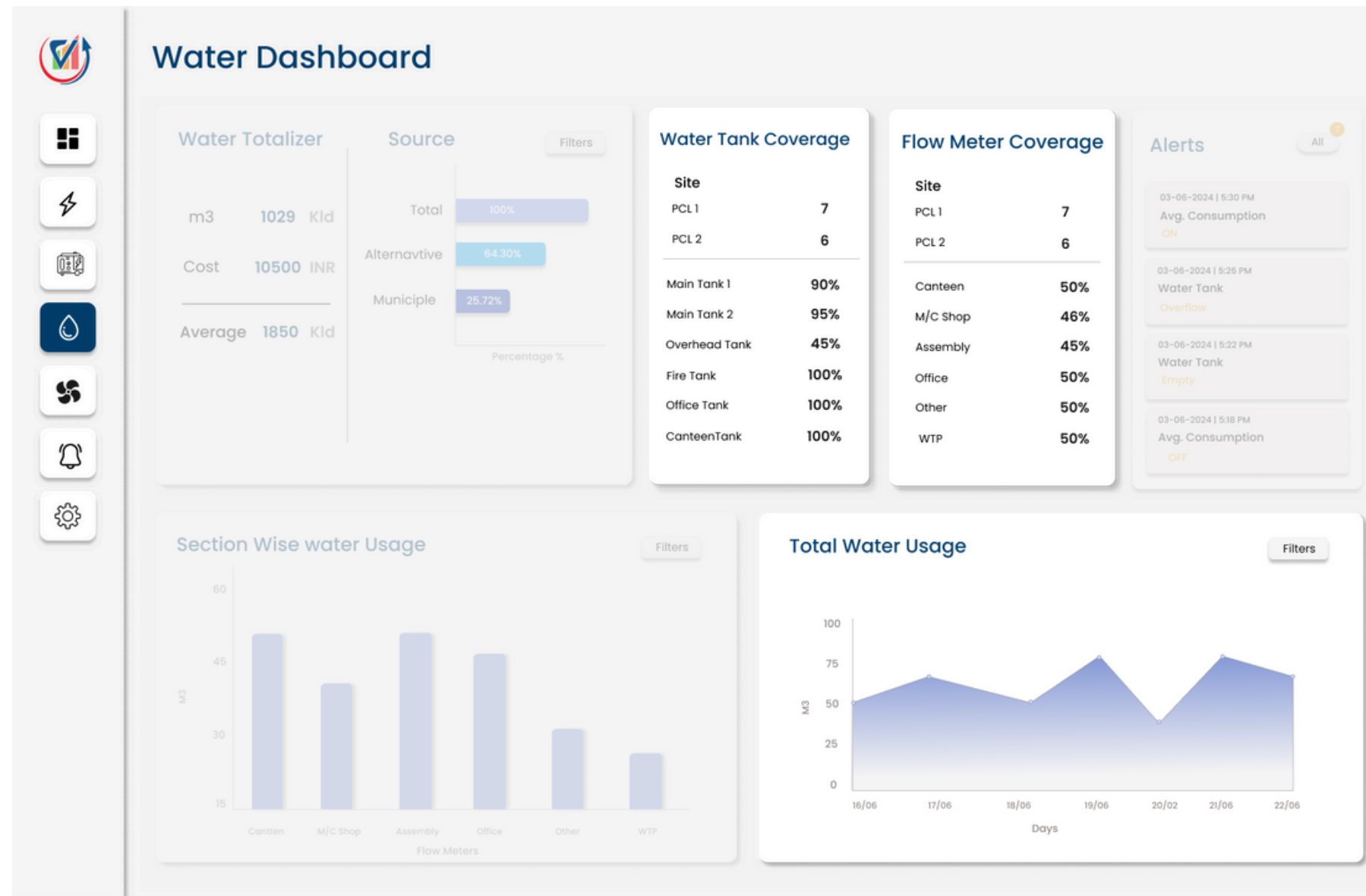
Water Totalizer:

- Displays total water usage in cubic meters (m³) and associated cost in INR.
- Shows average daily water usage in kiloliters per day (kld) to understand consumption patterns.

Source Breakdown:

- Allows users to view water consumption for today, yesterday, and month-to-date (MTD) from various sources.
- Indicates the overall percentage of water used.





Total Water Usage:

- Daily and monthly trends display a line graph of total water usage over a specified period, either daily or monthly.
- Helps identify usage patterns and trends for better planning and resource allocation.

Water Tank Coverage:

- Displays coverage and status of various water tanks.
- Coverage percentages indicate the fill level of each tank.
- Ensures water availability and prevents shortages and overflows.

Flow Meter Coverage:

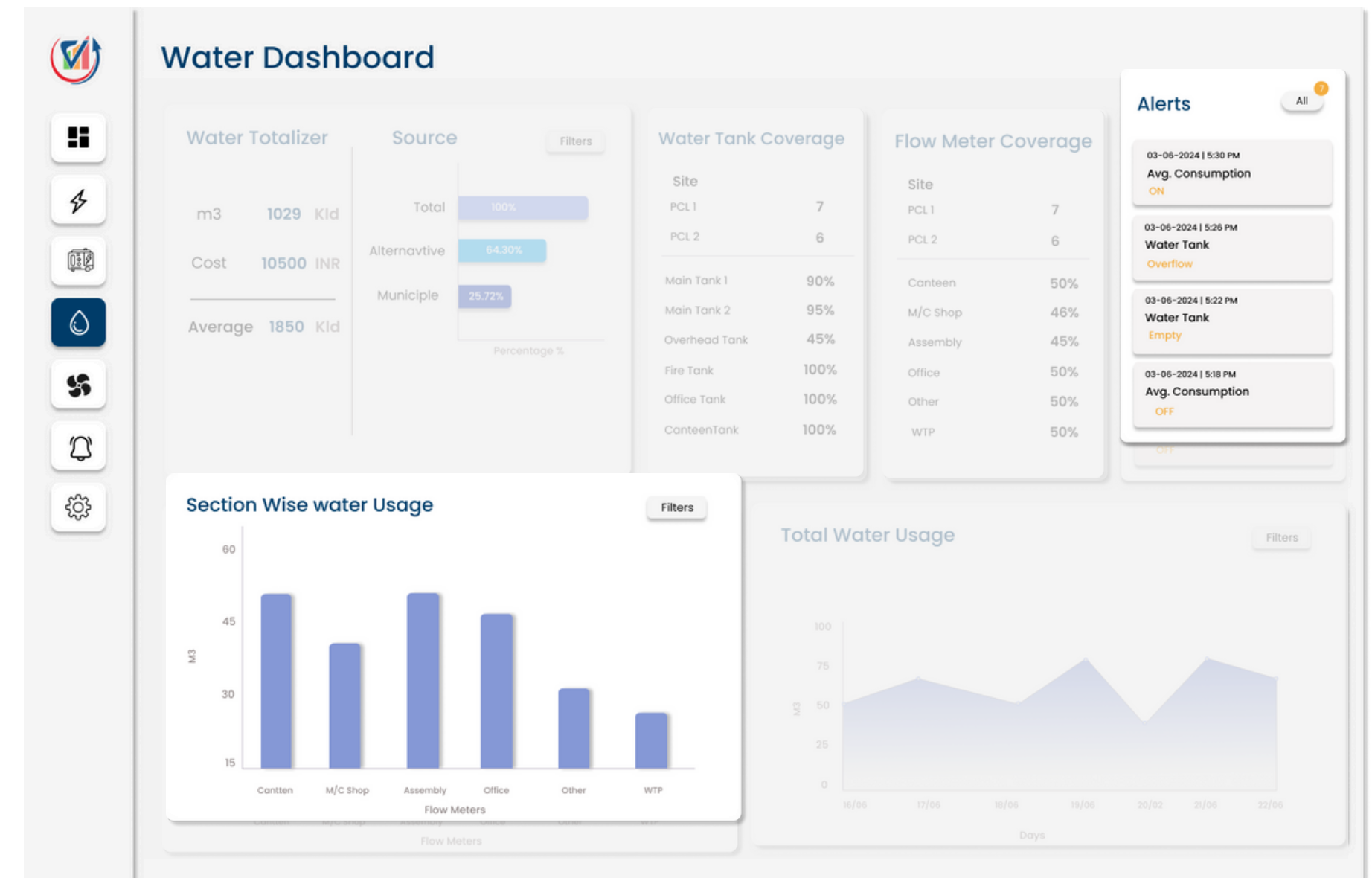
- Provides coverage information for various flow meters across different areas.
- Displays the number of flow meters in each area.
- Helps in detailed monitoring and accurate measurement of water flow.

Alerts:

- Displays real-time alerts for various water-related events.
- Alerts include average consumption status (ON/OFF), water tank status (Overflow, Empty), and specific equipment names.

Section-Wise Water Usage:

- Provides a visual breakdown of water usage across different sections for today, yesterday, and month-to-date (MTD).
- Helps identify areas with high water consumption and potential for savings.

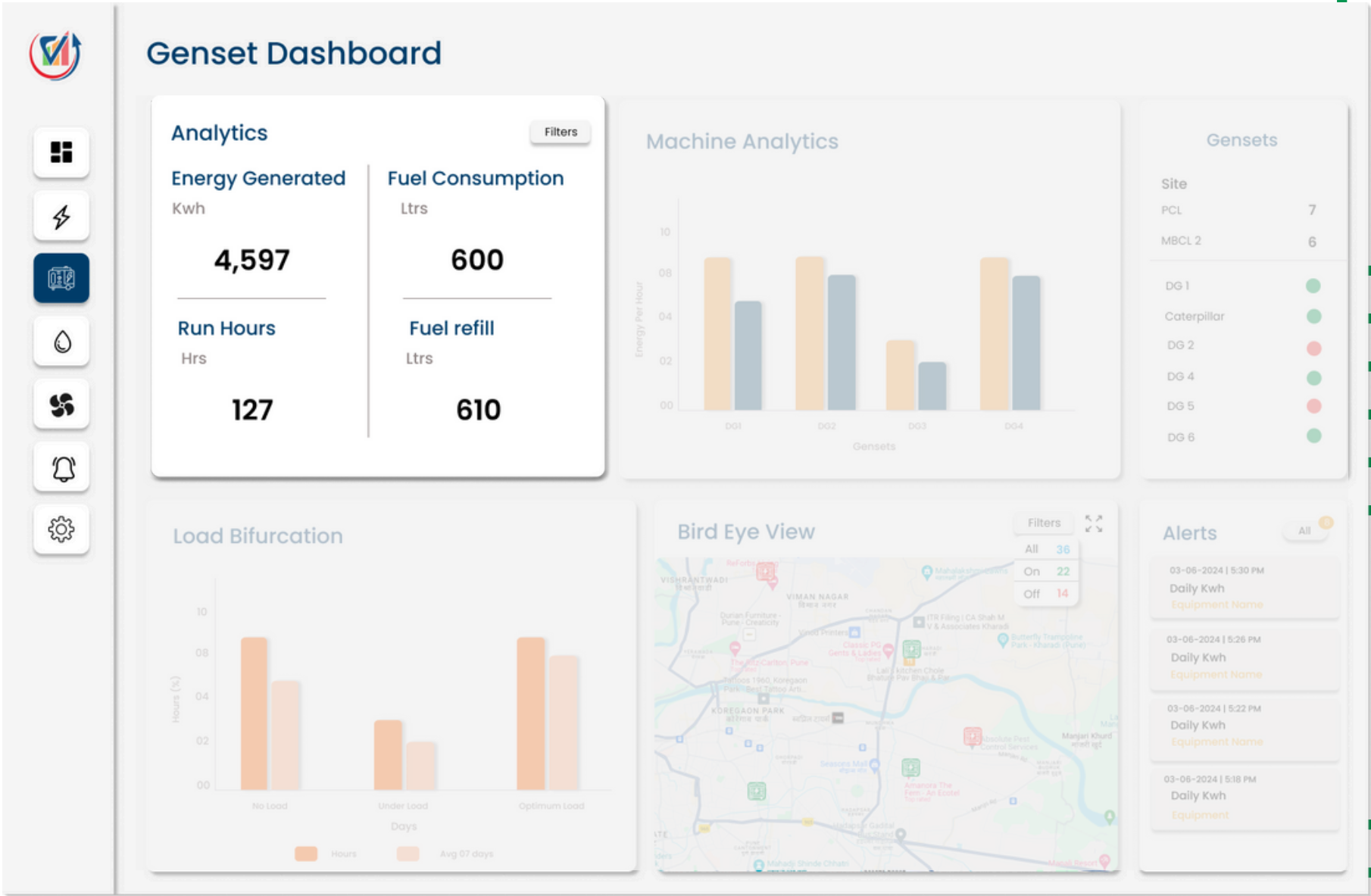


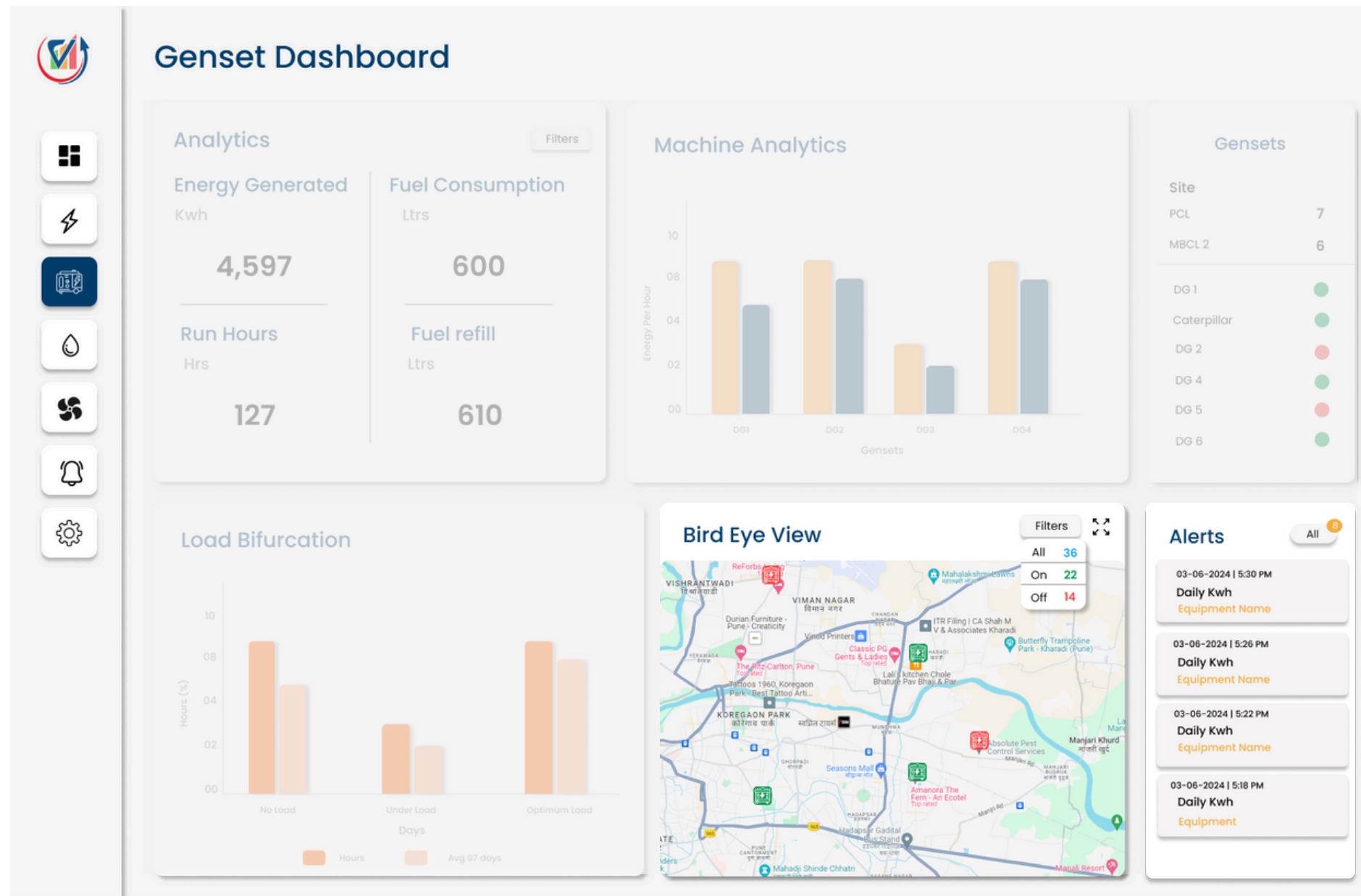
Genset Dashboard

The Genset Dashboard provides a comprehensive overview of generator set (genset) operations, enabling users to monitor and manage energy generation, fuel consumption, and overall performance effectively.

Analytics:

- **Energy Generated:** Displays total energy produced in kilowatt-hours (kWh).
- **Fuel Refill:** Monitors refilled fuel in liters, aiding in fuel logistics
- **Run Hours:** Indicates total genset operating hours for usage insights and maintenance scheduling.
- **Fuel Consumption:** Shows fuel used in liters (L), highlighting efficiency.





Geographical Overview

- Map view showing genset locations and operational status (On/Off).
- Quickly identifies geographical distribution and current status.
- Facilitates remote monitoring and management.

Alerts:

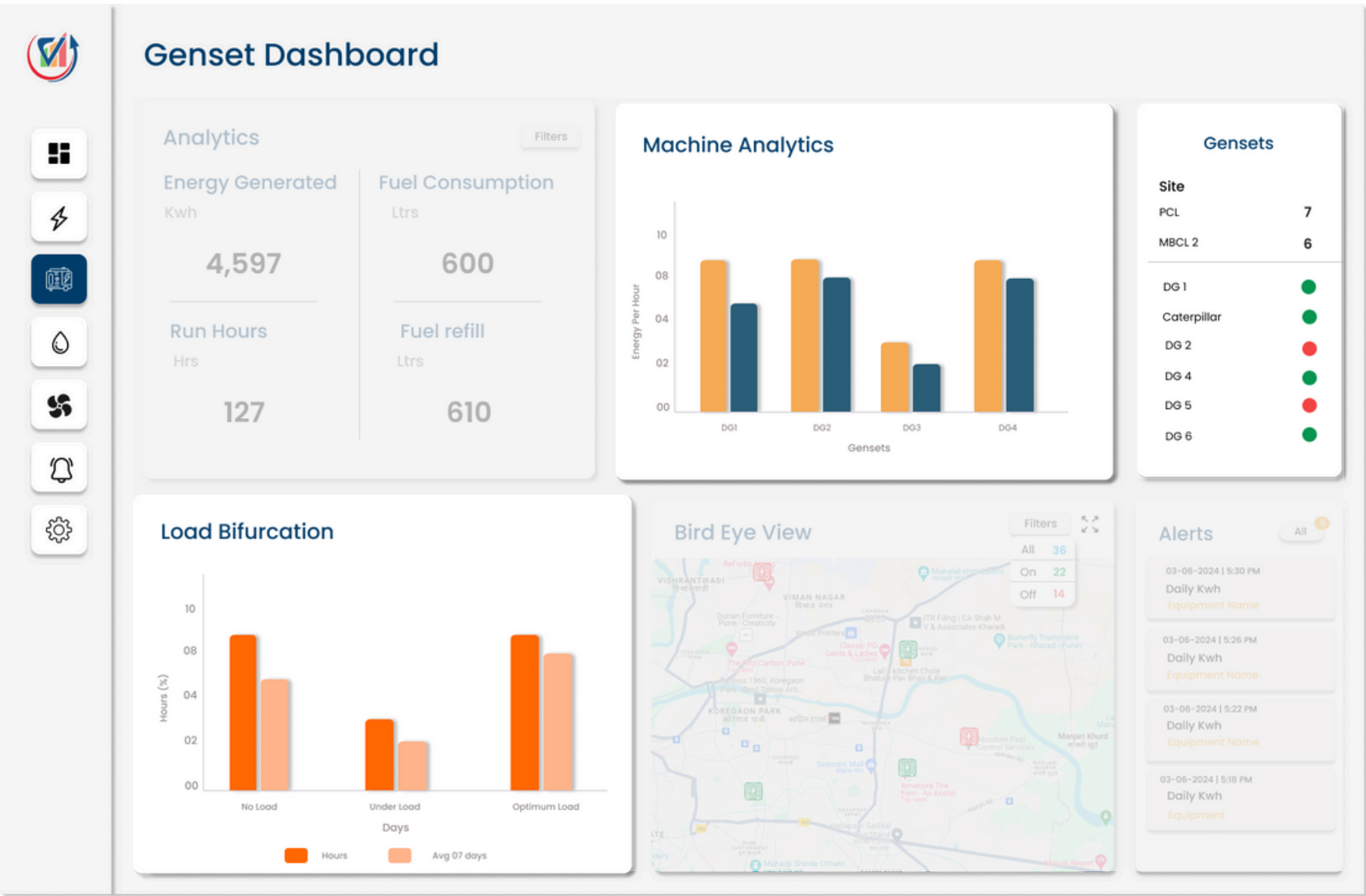
- Displays real-time alerts for various genset-related events.
- Includes issues like Daily kWh Equipment Name, MaxDemand, and other critical notifications.
- Ensures timely responses to operational issues, preventing downtime and ensuring reliable performance.

Machine Analytics:

- Visualizes key metrics for gensets via bar charts.
- Includes energy generation and fuel consumption rates for individual gensets.
- Helps compare performance and identify underperforming units.

Load Bifurcation:

- Shows load distribution categorized by No Load, Under Load, and Optimum Load conditions.
- Helps understand genset utilization and operational efficiency.
- Assists in load balancing and improving overall efficiency..



Genset Inventory:

- Provides a detailed list of all gensets, categorized by site and type.
- Aids in asset management and tracking each genset's status and performance.

THANK YOU

For your attention

Do you have questions or want to learn more? Reach out to us!

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